JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 93, NO. B10, PAGES 11,599–11,612, OCTOBER 10, 1988

Paleomagnetic Investigation of Some Volcanic Rocks From the McMurdo Volcanic Province, Antarctica

EDWARD A. MANKINEN AND ALLAN COX[1]

U.S. Geological Survey, Menlo Park, California

Paleomagnetic data for lava flows from sporadic but long-lived eruptions in the McMurdo Sound region are combined with previously published geologic and geochronologic data to determine the general eruptive sequence of the area. Lava flows in the Walcott Bay area were erupted during the Gauss Normal, Matuyama Reversed, and Brunhes Normal Polarity Chrons. The youngest flows on Black Island probably erupted near the boundary between the Gilbert and Gauss chrons. The most recent activity was concentrated on the volcanic edifices of Mounts Morning and Discovery and on Ross Island where Mount Erebus is currently active. Combining virtual geomagnetic poles calculated for 39 units sampled during this study with those of eight flows that were published previously yields a mean paleomagnetic pole at 87.3°N, 317.3°E (α_{95} = 6.3°). The ancient geomagnetic field dispersion about this mean pole is 23.5°, with upper and lower limits of 95% confidence equal to 27.4° and 20.5°, respectively. This value probably is a reasonable estimate of secular variation for the Antarctic continent during Pliocene and Pleistocene time.

INTRODUCTION

A major emphasis of paleomagnetic research within the U.S. Geological Survey (USGS) during the early 1960s was toward the development of a K-Ar polarity time scale. An excellent description of time scale research by the various laboratories and the investigators involved during this period was provided by *Glen* [1982]. Although less publicized, concurrent research at the USGS paleomagnetic laboratory in Menlo Park was directed toward the study of the secular variation of the geomagnetic field and particularly toward the dependence of the variation upon latitude. Collection of samples for secular variation studies between 1961 and 1967 were undertaken in Alaska and the western United States [*Cox and Doell*, 1964], France [*Doell*, 1970], the Galapagos Islands [*Cox and Doell*, 1964; *Cox*, 1971, also unpublished data, 1965], the Hawaiian Islands [*Cox and Doell*, 1964; *Doell and Cox*, 1965, 1971; *Doell*, 1969, 1972*a*, *b*; *Doell and Dalrymple*, 1973], Iceland [*Doell*, 1972*c*], and New Zealand [*Cox*, 1969*a*]. Paleomagnetic sampling of young volcanic rocks in the McMurdo Sound area of Antarctica for secular variation studies also was conducted by the current authors during the austral summer of 1965–1966. A value for the angular standard deviation of virtual geomagnetic poles (VGPs) from these rocks was shown on diagrams from *Doell* [1972*c*], *Doell and Dalrymple* [1973], *Doell and Cox* [1971, 1972], and the data shown for Antarctica in the latter publication were used in the analyses by *McElhinny and Merrill* [1975]. The paleomagnetic data for the individual flows, however, have never been published and it seems appropriate to do so in this special section.

The objectives of this study were the young volcanic rocks of Ross Island and surrounding areas of McMurdo Sound (Figure 1). The young volcanoes, including the active Mount Erebus, and their products have been known since the early days of Antarctic exploration because many of the expeditions had their headquarters in the area. A comprehensive description of the volcanic rocks collected by the scientific party of the British 1910–1913 "Terra Nova" expedition was provided by *Smith* [1954]. A formal nomenclature for the major stratigraphic units in this region was proposed by *Harrington* [1958], who designated the late Tertiary and Quaternary volcanic rocks of the area as the "McMurdo Volcanics." The volcanic rocks are similar petrographically and overlap in age, but the utility of assigning them to a single formation was questioned by *Hamilton* [1972], who referred to them collectively as belonging to the "Ross Sea petrologic province." This broad category was subdivided into three major subprovinces, the "McMurdo," "Hallett," and "Balleny" volcanic provinces, to characterize their different occurrences. Our investigation was confined to the McMurdo volcanic province, which is characterized by high subaerial volcanoes [*Hamilton*, 1972]. Within this province, Cenozoic volcanic rocks in some of the dry valleys on the mainland were mapped by *Blank et al.* [1963]; on Ross Island those at Cape Evans and Cape Royds were described by *Treves* [1962], and some on the southern tip of Hut Point Peninsula were described by *Wellman* [1964]. Our paleomagnetic sampling was guided by these publications and by a location map provided by W. Hamilton (written communication, 1965) showing where good exposures of the volcanic rocks could be seen. Geologic descriptions of some of the areas sampled have since been published by *Cole and Ewart* [1968] and *Cole et al.*, [1971], and K-Ar age determinations by *Armstrong et al.* [1968], *Treves* [1968], *Fleck et al.* [1972], *Forbes et al.* [1974], and *Armstrong* [1978] establish an age range for the volcanic rocks. *Armstrong* [1978] also collected oriented hand samples from many of the flows to later determine their magnetic polarity (using a portable flux gate magnetometer) as a test of the validity of the K-Ar determinations. Oriented paleomagnetic cores were obtained from some of the volcanic rocks on Ross Island by *Kyle and Treves* [1974*a*] and *Funaki* [1979, 1984], and paleomagnetic studies were made on a core obtained on Ross Island by the Dry Valley Drilling Project [*McMahon and Spall*, 1974]. Paleomagnetic results from Cenozoic volcanic rocks in the Hallet volcanic province near Capes Hallett and Adare, northern Victoria Land, were reported by *Turnbull* [1959] and *Del-*

[1]Deceased January 27, 1987.

This paper is not subject to U.S. copyright. Published in 1988 by the American Geophysical Union.

Paper number 7B3044.

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

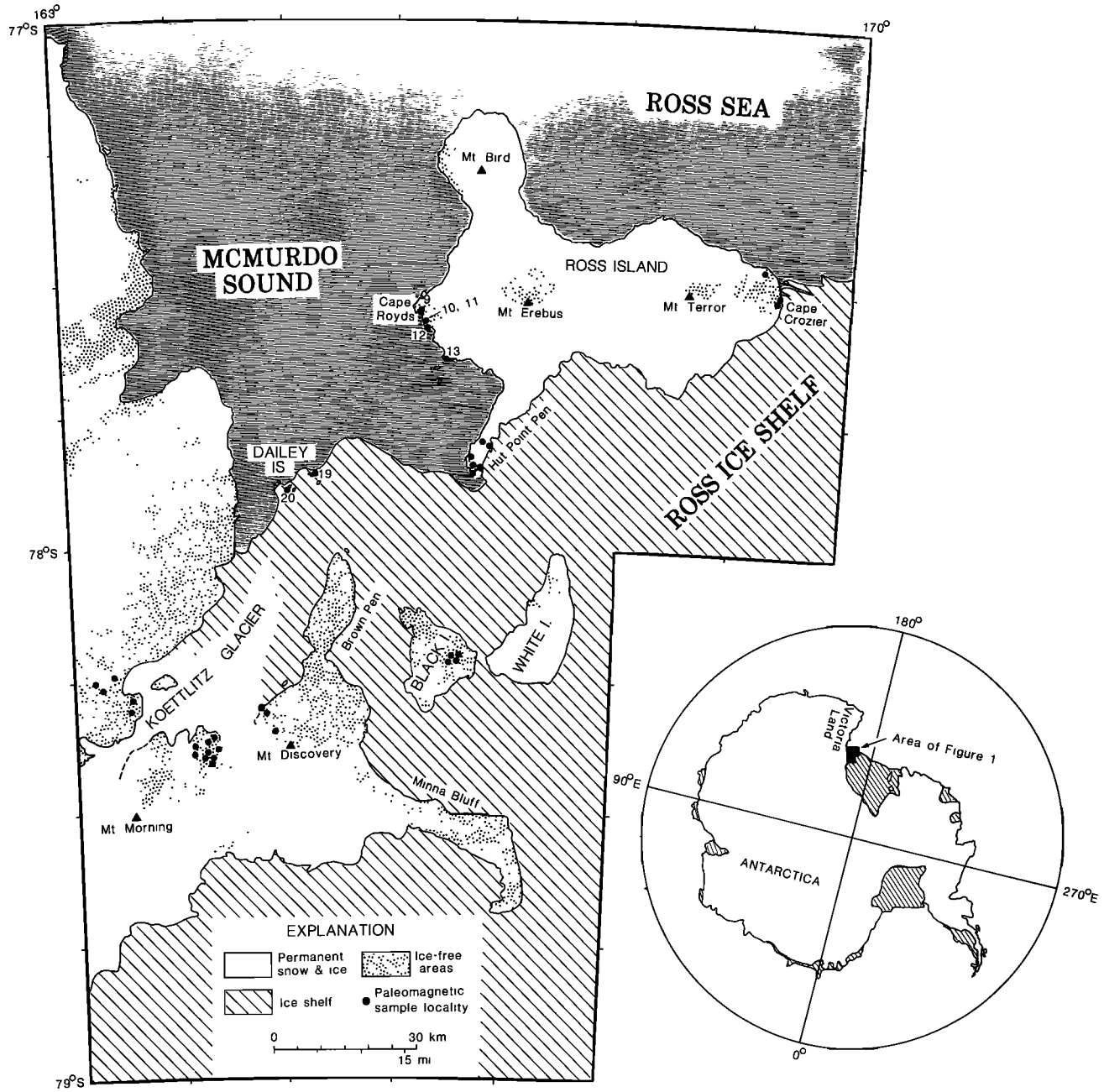


Fig. 1. Map of the McMurdo Sound region showing the general distribution of paleomagnetic sampling sites.

isle [1983] and from those in Marie Byrd Land by *Scharon et al.* [1969].

METHODS AND RESULTS

Paleomagnetic samples were obtained by use of a portable core drill. Placing pressurized cans containing very hot water in a knapsack with sample bags for insulation provided drilling water for a few hours and made it unnecessary to use an antifreeze liquid. A minor problem with freezing of the water shut-off valve was occasionally encountered but readily corrected with the use of a propane torch. Cores were oriented with a magnetic compass and corrections applied to the measured azimuths were obtained by backsighting to one or more distant landmarks. Antarctica 1:250,000 Reconnaissance Series maps published by the U.S. Geological Survey were used for location. The plane table maps of *Debenham* [1921] and *Smith* [1954] also proved useful for location in some areas of Ross Island. Because we intended to sample mainly those lava flows that erupted during the Brunhes Normal-Polarity Chron, the polarity of an outcrop was first measured with a portable flux gate magnetometer. We found that the flux gate magnetometer would remain operational even in extreme cold if it was worn under one's parka except when in use. Some lava flows were not sampled in detail, but the polarity of four separate samples was measured in the field. If all yielded consistent results, one sample was carefully oriented and returned to the laboratory to be cored and measured. In a few instances, only field determination of polarity were made. A total of 303 oriented cores was obtained from 39 localities, and oriented hand samples were collected from an additional 11 localities, all of which are shown in Figure 1. None of the units sampled have been tectonically disrupted. Each core was cut into two or more specimens, and the remanent mag-

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

netization of one specimen from each core was measured using a spinner magnetometer. A representative specimen from each locality was subjected to alternating field (af) demagnetization using a tumbling demagnetizer to test the stability of remanence.

It became evident very early in our field determinations of magnetic polarity that extremely young looking volcanic features could have reversed polarity, which indicated that processes of erosion and weathering must proceed very slowly in that environment [*Cox*, 1966]. *Wellman* [1964] had remarked previously that topographic features could be much older than they appear if judged by normal standards. This finding also made it possible that all of the normal-polarity flows sampled might not be of Brunhes age; a possibility that was reinforced as K-Ar age determinations became available. On the other hand, the extremely arid environment ensured that the remanence of these volcanic rocks was essentially undisturbed. Very few of the samples analyzed showed any change in remanence direction during af demagnetization. One specimen from each core collected from the Observation Hill trachyte (site 1) was demagnetized at 50 Oe to test the usefulness of magnetic cleaning. The mean direction changed by only 1.9°, and the associated statistics of the mean improved slightly (k from 274 to 343 and α_{95} from 3.4° to 3.0°). The pilot af demagnetization experiments indicated the need for magnetic cleaning for only three sites near Mount Morning (sites 33, 41, and 42) and one site near Walcott Glacier (site 49). Cleaning at either 25 or 50 Oe changed the mean directions of the Mount Morning sites by 1.2°, 1.9°, and 4.5°. The most significant change was for the Walcott Glacier site whose mean direction changed by 10.2°. Consequently, only the data for these five sites among those given in Table 1 have been magnetically cleaned. It is noted, also, that magnetic cleaning at 150 Oe of samples from 12 of the 13 sites reported by *Funaki* [1984] changed the site mean directions by 2.0° or less and had a negligible effect on their statistical parameters. The remaining site contained a normal-polarity overprint (removed by 350 Oe) upon a primary reversed direction. Thus it appears that magnetic cleaning rarely has any effect on the remanence directions of any of these lava flows.

The rocks sampled in the various areas around McMurdo Sound and the results obtained will be described in the sections that follow. It should be pointed out that any rock names used should be only considered to be "field descriptions." For example, *Smith* [1954] has shown that many of the basalts in the region are undersaturated and should more accurately be described as basanitic basalts, or even basanites. Similarly, the trachytic rocks were found to range from trachytoid phonolites to phonolitic trachytes. Most of the potassium-argon ages that are cited will vary slightly from those originally reported because we have corrected them to conform with currently accepted abundance and decay constants [*Steiger and Jäger*, 1977].

Cole and Ewart [1968] identified a stratigraphic sequence of volcanic rocks on Black Island that seemed to be repeated in different areas around McMurdo Sound [*Cole and Ewart*, 1968; *Cole et al.*, 1971]. Four formational names were proposed: Trachyte Hill Formation (youngest), Nubian Basalt, Aurora Trachyte, and the Melania Basalt (oldest). Although the stratigraphic sequence of alternating basalt and trachyte is present in many areas, the age determinations of *Armstrong* [1978] show that the different volcanic centers have unique ages. These formal designations, therefore, are not stratigraphically useful over much of the area, but the terms will be used wherever applicable in this report in order to correlate our paleomagnetic sampling to the mapped geology.

Ross Island

Hut Point Peninsula (sites 1–8). Many young lava flows are exposed on Hut Point Peninsula within easy access of the United States' McMurdo Station and New Zealand's Scott Base (Figure 2). Most of the rocks on the peninsula are basalts except for the trachyte that forms Observation Hill and the volcanic breccia at Castle Rock and Boulder Cones. Several of these units have now been sampled for paleomagnetic study by us, by *Kyle and Treves* [1974*a*], and by *Funaki* [1979, 1984]. *Kyle and Treves* [1974*a*] note, however, that instrumental and field orientation errors for their samples may be as much as ±20° for declination and ±10° for inclination, so their results should be considered to be satisfactory only for polarity determination. Three holes also have been drilled and cored as part of the Dry Valley Drilling Project (DVDP) [*Treves and Kyle*, 1973; *Kyle and Treves*, 1974*b*; *Kyle*, 1978, 1981], and paleomagnetic measurements made on all units encountered in DVDP hole 2 were reported by *McMahon and Spall* [1974].

Our paleomagnetic sample localities on Hut Point Peninsula are shown in Figure 2, and site mean directions of magnetization of the units sampled are shown in Figure 3. Flux gate polarity measurements showed that all of the trachytes composing Observation Hill are reversely magnetized [*Cox*, 1966] and probably erupted during the Matuyama Reversed-Polarity Chron. This conclusion was corroborated by K-Ar age determinations on samples from three localities, on the northwest side and on the summit, that average 1.25 ± 0.03 Ma [*Forbes et al.*, 1974]. Paleomagnetic samples were obtained from a flow on a prominent rib on the north side of Observation Hill (site 1). *Kyle and Treves* [1974*a*] and *Funaki* [1979, 1984] also sampled some of these trachytes and found reversed polarities. The remanence directions determined for the rocks at our site 1 and *Funaki*'s [1984] site 2 are statistically different at the 95% confidence level using the method of *McFadden and Jones* [1981], which indicates that two separate flows were sampled.

Paleomagnetic samples were obtained from basalt flows at two localities south and southeast of Crater Hill. Site 2 is in the uppermost of several thin flows separated by scoriaceous layers on the south flank of Crater Hill, and site 3 is in an approximately 10-m-thick flow at the base of the cliff northeast of Scott Base. *Kyle and Treves* [1974*a*] obtained paleomagnetic samples from a site near Scott Base, and *Funaki* [1979, 1984] sampled three sites within the Crater Hill sequence. All six sites proved to be of normal polarity. Results from the three sites (3, 7, and 8) given by *Funaki* [1984] do not differ significantly, partly because of the large uncertainty limits about the directions for sites 7 and 8 (11.7° and 12.8°, respectively). The remanence directions at our sites 2 and 3 differ from each other and from the direction at Funaki's site 3. Thus eruption of flows during at least three separate periods is indicated.

Basalt forming a series of irregular dikelike bodies was sampled along the top of Fortress Rock (site 4). *Funaki* [1984] also sampled one of the basalts at Fortress Rock (his site 1), which he considered as belonging to the Twin Crater sequence. Tentatively included in the Twin Crater sequence [*Treves and Kyle*, 1973; *Kyle and Treves*, 1974*a*] are flows at Black Knob, First and Second Craters, Hut Point, and the lavas along Arrival Heights, one of which we sampled (site 6).

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Site-Average Paleomagnetic Data for Volcanic Rocks From the McMurdo Sound Area, Antarctica

Map	Site Latitude °N	Site Longitude °E	N	Inc., deg	Dec., deg	R	k	α_{95}	J_{nrm}	VGP Latitude °N	VGP Longitude °E
1	−77.85	166.64	8	77.0	255.9	7.9798	346	3.0	2.11	−65.3	089.2
2	−77.85	166.69	8	−82.1	334.3	7.9894	659	2.2	1.94	83.1	092.1
3	−77.84	166.76	8	−85.4	349.9	7.9890	637	2.2	5.02	86.5	014.6
4	−77.84	166.70	7	−85.7	002.4	6.9881	506	2.7	3.70	86.4	341.1
5	−77.84	166.67	1	70	066				8.15		
6	−77.83	166.67	8	−78.6	009.8	7.9775	311	3.1	4.61	79.7	187.6
7	−77.80	166.72	8	−69.3	165.8	7.9749	278	3.3	17.1	41.0	335.4
8	−77.80	166.83	8	−78.4	042.5	7.9704	236	3.6	10.5	74.4	239.8
9	−77.55	166.20	8	−84.1	244.0	7.9878	575	2.3	1.82	69.6	017.5
10	−77.57	166.23	8	−84.1	341.0	7.9796	344	3.0	2.63	86.0	056.5
11	−77.57	166.23	8	−80.0	326.8	7.9890	639	2.2	2.79	78.8	097.1
12	−77.58	166.23	6	74.7	130.8	5.9666	155	5.4	9.54	−67.5	274.2
13	−77.64	166.41	8	−75.2	111.3	7.9718	248	3.5	2.66	55.8	295.6
14	−77.46	169.23	8	−82.8	004.9	7.9815	378	2.9	10.1	88.0	206.7
15	−77.47	169.23	8	84.2	170.5	7.9937	1105	1.7	1.04	−87.8	226.9
16	−77.51	169.38	6	69.1	152.6	5.9573	117	6.2	1.38	−63.2	311.2
17	−77.52	169.35	1	78	182				9.93		
18	−77.52	169.35	1	64	152				1.14		
19	−77.86	165.23	8	−80.0	144.0	7.9904	726	2.1	14.3	59.9	322.3
20	−77.88	165.02	8	−77.3	129.2	7.9765	298	3.2	16.0	56.8	309.5
21	−78.21	166.49	8	79.2	326.9	7.9799	348	3.0	2.99	−58.6	144.5
22	−78.20	166.54	2	45	036				0.39		
23	−78.20	166.58	1	−68	256				7.00		
24	−78.21	166.57	1	−78	050				10.6		
25	−78.21	166.57	1	76	136				6.91		
26	−78.21	166.57	8	−56.7	013.2	7.9734	263	3.4	2.49	48.7	182.6
27	−78.21	166.56	1	−75	059				5.66		
28	−78.29	164.73	6	−84.0	103.5	5.9735	189	4.9	4.42	71.5	305.6
29	−78.31	164.80	8	−80.2	005.8	7.9785	325	3.1	12.6	82.5	179.3
30	−78.34	164.87	8	68.5	247.2	7.9720	250	3.5	7.36	−54.9	067.3
31	−78.34	164.28	8	−85.8	258.5	7.9855	482	2.5	3.27	74.3	016.4
32	−78.36	164.30	8	−73.7	268.0	7.9752	282	3.3	11.2	57.4	053.5
33	−78.38	164.34	8	−76.5	018.4	7.9647	198	4.0	4.14	75.0	196.1
34	−78.39	164.27	7	−83.1	271.6	6.9883	513	2.7	17.9	72.4	035.6
35	−78.39	164.23	8	−83.7	292.1	7.9775	312	3.1	7.39	76.6	044.0
36	−78.39	164.27	8	−84.9	002.1	7.9622	185	4.1	8.62	88.5	330.1
37	−78.40	164.27	8	82.1	209.7	7.9842	443	2.6	8.02	−82.1	059.5
38	−78.40	164.21	8	−76.7	297.2	7.9765	298	3.2	9.86	67.7	075.7
39	−78.39	164.21	8	−86.4	266.3	7.9596	173	4.2	5.41	76.0	015.3
40	−78.39	164.20	8	−85.3	191.8	7.9539	152	4.5	6.60	69.2	349.5
41	−78.39	164.12	8	−78.6	266.5	7.9616	182	4.1	8.57	64.7	044.8
42	−78.37	164.12	8	−80.5	337.0	7.9344	107	5.4	0.92	81.0	111.8
43	−78.37	164.24	8	−88.5	265.6	7.9513	144	4.6	12.1	77.8	358.6
44	−78.36	164.26	7	−75.6	332.2	6.9936	940	2.0	4.47	72.4	119.7
45	−78.28	163.55	1	65	203				4.96		
46	−78.31	163.52	1	68	012				7.61		
47	−78.26	163.25	1	−33	008				0.41		
48	−78.24	163.36	8	−59.8	074.8	7.9512	143	4.6	3.29	42.7	248.6
49	−78.24	163.36	8	75.5	164.7	7.9363	110	5.3	4.49	−73.6	317.8
50	−78.25	163.22	8	71.6	189.4	7.9841	439	2.7	5.04	−67.9	357.2

N, number of samples used in the analysis; Inc., Dec., inclination, positive downward, and declination, east of north, respectively, of remanence vector; k, precision parameter ($N - 1/N - R$); α_{95}, 95% confidence cone about average direction; J_{nrm}, geometric mean intensity of natural remanent magnetization (NRM) direction ($\times 10^{-3}$ emu/cm^3).

The lava flows in this sequence are among the youngest rocks of Hut Point Peninsula. *Armstrong* [1978] obtained ages of 0.44 ± 0.07 Ma for basalt from Black Knob and 0.59 ± 0.03 Ma for basalt immediately southwest of Black Knob. *Funaki* [1979, 1984] obtained samples from Black Knob (his site 11), from Second Crater (his site 10), and from near Scott's Hut at Hut Point (his site 5). *Kyle and Treves* [1974*a*] also sampled Second Crater and Twin Crater. An oriented hand sample from our site 5, which is in the same general area, is reversely magnetized. The site sampled had just been exposed due to excavation by U.S. Navy personnel and may be part of the old crater rim shown in the sketch by *Wellman* [1964]. Except for the reversed-polarity direction of this unit and a possible intermediate-polarity direction recorded by the Twin Crater flow sampled by *Kyle and Treves* [1974*a*], all flows sampled in this immediate area are of normal polarity. Results from the two Fortress Rock localities are statistically indistinguishable and indicate that the same unit or products of the same eruptive sequence were sampled. Flows sampled by *Funaki* [1979, 1984] near Scott's Hut and at Second Crater have remanence directions that are statistically the same, and both differ from that at our Arrival Heights site.

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

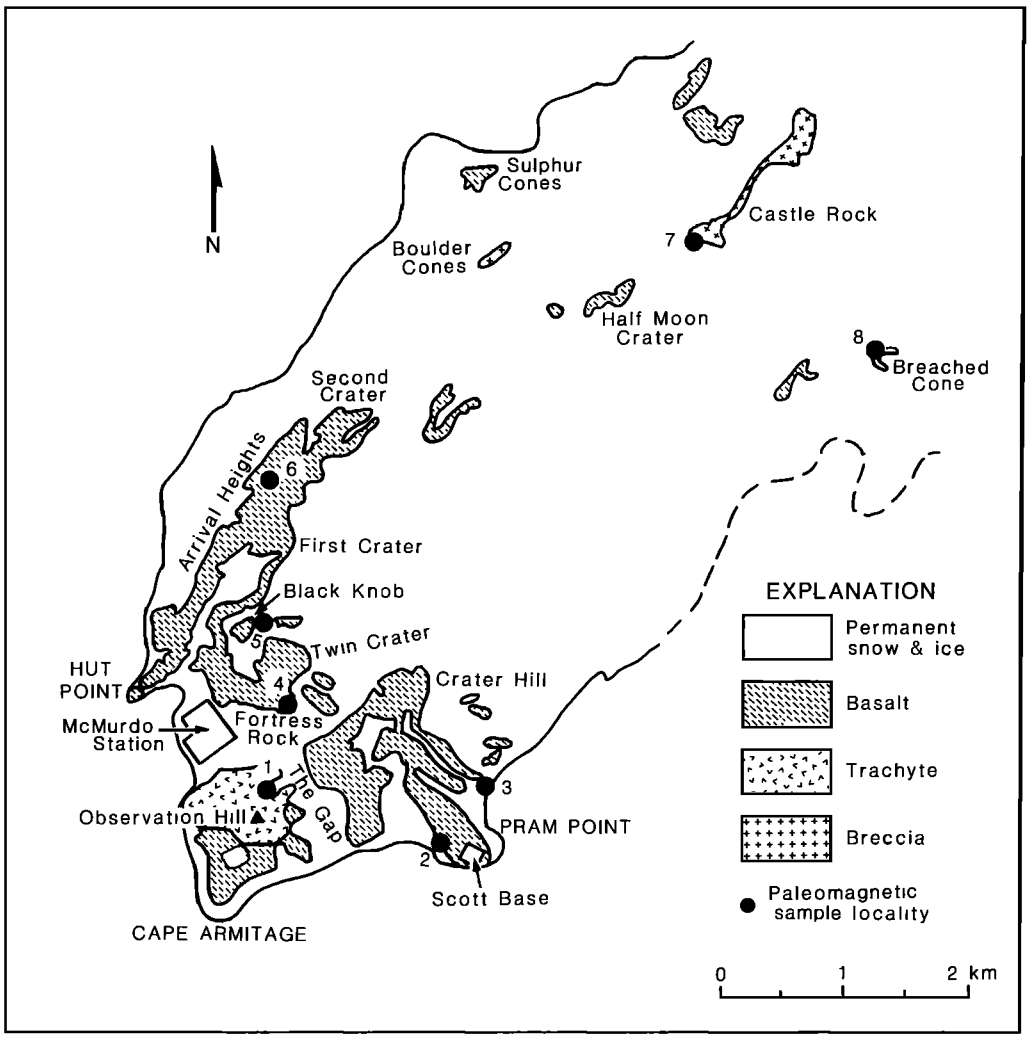


Fig. 2. Map of the tip of Hut Point Peninsula showing the location of volcanic units sampled during this study. Outcrop distribution and petrology is modified from *Cole et al.* [1971].

Farther to the northeast on Hut Point Peninsula, *Kyle and Treves* [1974*a*] sampled a hawaiite flow at Half Moon Crater that proved to be of normal polarity and to have an age of 1.0 ± 0.2 Ma [*Kyle and Treves*, 1974*a*; *Armstrong*, 1978]. The same unit was sampled in greater detail by *Funaki* [1979, 1984]. We obtained samples from a dike (site 7) that intrudes the breccia of Castle Rock approximately 0.5 km northeast of Half Moon Crater and from a small basalt flow (site 8) in Breached Cone approximately 1.5 km east of Castle Rock. Both units possess normal polarity and the dike yields an age (1.1 ± 0.4 Ma [*Kyle and Treves*, 1974*a*]) that is virtually the same as that determined for the Half Moon Crater flow. The proximity and age of these two units are suggestive of emplacement at about the same time which, when uncertainties in the age determinations are considered, was most probably during either the Jaramillo or Cobb Mountain Normal-Polarity Subchrons. Markedly different remanence directions, however, indicate that the two units differ in age by at least some tens or hundreds of years. The remanence direction of the dike yields a virtual geomagnetic pole (VGP) that has a latitude of 41.0°. Such a low VGP latitude commonly is considered to represent a transitional polarity, but it possibly may be near the limits of normal secular variation at high-latitude sampling sites. The remanence direction of the Breached Cone flow differs from both the Half Moon Crater and Castle Rock units, and without independent age information it is not certain if all three units were emplaced during the same polarity interval. The dike, whose top is brecciated, is thought to have fed lava to the ice contact zone where the palagonitic breccia of Castle Rock was formed [*Kyle and Treves*, 1974*a*]. However, the difference in remanence directions of the dike (Table 1) and the matrix samples of the breccia [*Funaki*, 1979, 1984] indicate that this assumption may be incorrect. Alternatively, the three samples collected by Funaki may not be entirely representative of the true remanence direction of the breccia, or perhaps the geomagnetic field was not accurately recorded by the unit.

The inferred succession of surficial volcanic units as summarized by *Kyle and Treves* [1974*a*] is, from youngest to oldest, the (1) Twin Crater, (2) Half Moon Crater, (3) Castle Rock, (4) Observation Hill, and (5) Crater Hill sequences. Thus it appears that very few of the paleomagnetically sampled flows actually erupted during the Brunhes Chron. Uncertainties in the ages of the Half Moon Crater and Castle Rock sequences do not preclude eruption during the Brunhes but, as noted above, a slightly older period of activity may be more likely. If the Crater Hill sequence is older than the Observation Hill flows, it could have erupted no later than the Olduvai

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

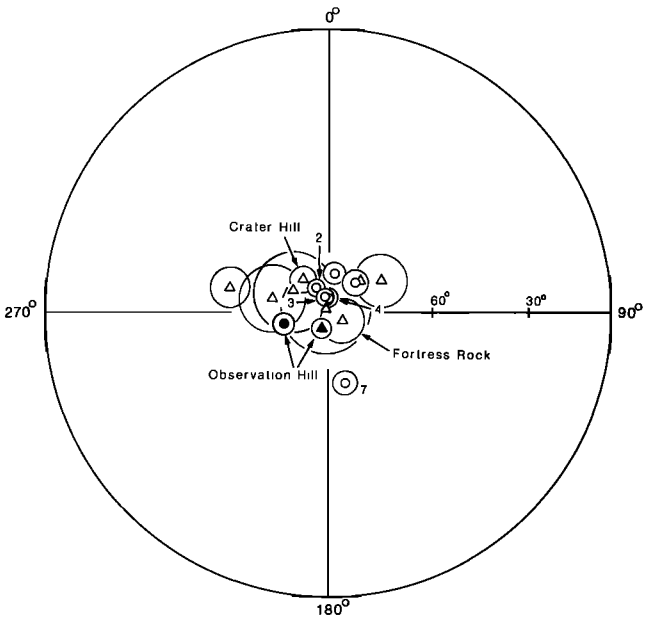


Fig. 3. Average remanent magnetization directions for units sampled on Hut Point Peninsula. Circles, data from this study; triangles, data from *Funaki* [1984]; ovals, 95% confidence limits about mean directions. Open symbols, directions on upper hemisphere (normal polarity) of equal-area projection; solid symbols, directions on lower hemisphere (reversed polarity).

Normal-Polarity Subchron which would place the age of three flows between 1.87 and 1.67 Ma [*Mankinen and Dalrymple*, 1979]. Such an age would make the Crater Hill sequence at least 0.4 m.y. older than both the surface volcanics that have been dated and the sequence of flows and pyroclastic units encountered in the three DVDP drill holes. All 15 units overlying the basal pyroclastic sequence in DVDP hole 2 are reversely magnetized [*McMahon and Spall*, 1974] and K-Ar age determinations on four DVDP core samples average 1.26 ± 0.13 Ma [*Kyle et al.*, 1978]. *Kyle et al.* [1978] suggest that the entire drill hole sequence erupted in less than 200,000 years. Eruption of the Crater Hill sequence during the Olduvai, therefore, seems unlikely especially because the hyaloclastite at the bottom of the three drill cores was interpreted to record a period of submarine volcanism which produced a pedestal upon which the surficial flows accumulated [*Kyle and Treves*, 1974*b*]. The Crater Hill sequence was determined to be the oldest in the area because trachytes of the Observation Hill sequence were found to overlie Crater Hill flows at The Gap and at Cape Armitage [*Kyle and Treves*, 1974*a*]. It would seem more likely that the trachyte is intercalated in the Crater Hill sequence lavas and overlies only the oldest flows, which may be close in age to the trachyte itself. Corroborating this possibility are the approximately 1.26 Ma ages determined for basalt flows in the DVDP drill cores [*Kyle et al.*, 1978] and the reversed polarity of a basalt flow underlying the trachyte at Cape Armitage [*Funaki*, 1979, 1984]. The normal-polarity Crater Hill flows then could be younger than the Observation Hill trachytes and correlatives of some of the other surficial normal-polarity flows. Until additional age information becomes available, it appears that most of the surface volcanic units near the tip of Hut Point Peninsula erupted during the latter part of the Matuyama Chron and the beginning of the Brunhes.

Capes Royds, Barne, and Evans (sites 9–13). These three capes on the west side of Ross Island are composed mainly of kenyte [*Smith*, 1954; *Treves*, 1962] or anorthoclase trachyte [*Boudette and Ford*, 1966] flows from Mount Erebus, with some basaltic vents localized at Cape Barne [*Smith*, 1954]. A considerable portion of the surface area at Cape Royds and Barne is mantled by morainal material. Moraines containing erratics of mainland rocks are present as are others consisting of locally derived volcanic rocks. Of the latter, many occur as terraced moraines [*Debenham*, 1921] producing a stepped appearance to the topography. Debenham also described the ridges of gravel and snow that were produced at Cape Evans by the strong prevailing winds. Three major kenyte flows appear to be present at Cape Royds [*Treves*, 1962], and *Treves* [1968] reported a K-Ar age of 0.70 ± 0.14 Ma for the second lowest flow. We sampled one of the flows (site 9) along the east side of Backdoor Bay, and *Funaki* [1984] obtained paleomagnetic samples from two flows north of Shackleton's Hut, both of which are north of our locality. The remanence directions for localities Royds A and Royds B [*Funaki*, 1984] are distinct, but those at Royds A and at our site 9 are statistically indistinguishable. Because these two localities are probably less than 2000 m apart, it is possible that the same flow was sampled. South of Backdoor Bay, two superimposed kenyte flows were sampled (sites 10 and 11) about 300 m west of the largest of the frozen lakes north of Cape Barne. Also sampled was a basaltic dike (site 12) near the consolidated vent of a denuded cone that is known as the "Cape Barne pillar." The kenytes possess normal-polarity directions that, although they are statistically distinguishable, are similar enough to indicate that they do not differ greatly in age. The 0.97 ± 0.05 Ma age obtained by *Armstrong* [1978] for a kenyte from this area indicates that they were probably erupted during the Jaramillo Subchron. Controversy exists as to whether the basalts at Cape Barne are older or younger than the kenytes in their vicinity [*Smith*, 1954] and the uncertainty about the 0.8 ± 0.2 Ma age [*Armstrong*, 1978] of one of the basalts is too large to distinguish between the alternatives. The reversed-polarity direction of the dike at site 12, however, does indicate eruption during the Matuyama Chron. The magnetic polarities and available age determinations indicate that the flows on this part of Ross Island erupted during the latter part of the Matuyama Chron and the beginning of the Brunhes. The volcanism in this area was therefore partly contemporaneous with that on Hut Point Peninsula.

South of the Barne Glacier, Cape Evans consists of at least two kenyte flows [*Treves*, 1962]. The tip of Cape Evans appears to consist of one thick kenyte flow which we sampled at the base of an approximately 12-m-high cliff on the south side of the cape (site 13). This flow has a normal-polarity direction as does a flow that is exposed along the beach approximately 300 m northeast of Scott's Hut. The polarity of this latter flow was determined by measuring four separate samples with a flux gate magnetometer. No absolute ages have been determined for the rocks at this locality.

Cape Crozier (sites 14–18). An extensive area of ice-free terrain occurs at Cape Crozier where a number of basalt and trachyte cones and flows are exposed. Thick deposits of morainal material containing granitic erratics mantles much of the area between the rock outcrops. Paleomagnetic sample localities are shown in Figure 4 along with the main volcanic outcrops as mapped by *Cole et al.* [1971]. A flow (site 14) within the large area of olivine-augite basalt west of the Dis-

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

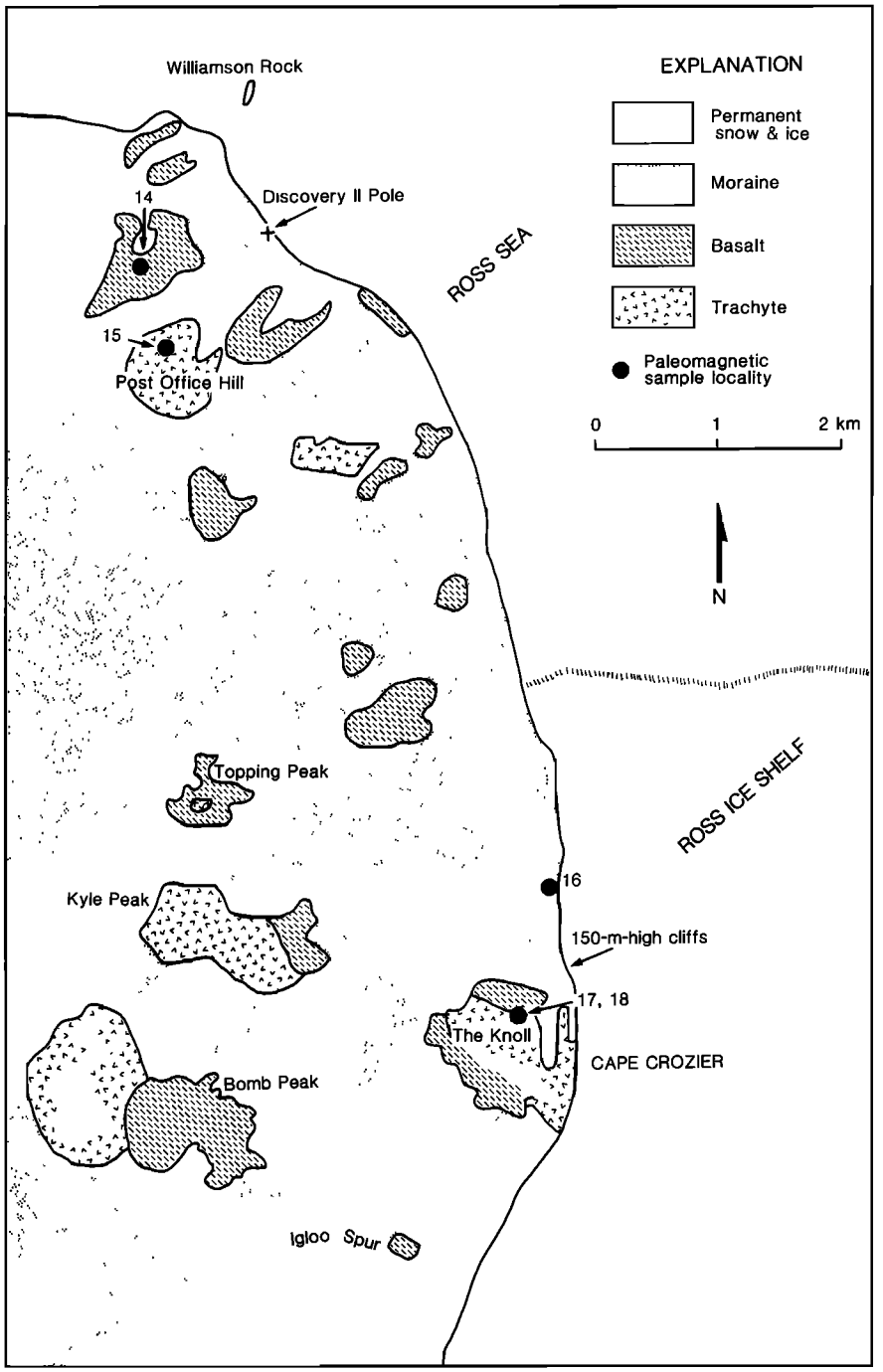


Fig. 4. Map of Cape Crozier showing location of volcanic units sampled. Outcrop distribution and petrology modified from *Cole et al.* [1971].

covery II Pole (Figure 4) was the only normal-polarity lava found in our reconnaissance of the vicinity. Flows in this area are considered to be part of the Melania basalt by *Cole et al.* [1971] as is a dike (site 16) that was sampled on the coastal cliffs north of The Knoll. A red scoriaceous basalt capping a ridge in the area of Melania basalt approximately 1 km northeast of Topping Peak proved to be of reversed polarity on the basis of flux gate polarity measurement of four samples. Rocks correlated with the Aurora trachyte by *Cole et al.* [1971] were sampled at Post Office Hill (site 15) and at The Knoll (site 18), both of which were reversely magnetized. A Nubian basalt flow (site 17) capping the trachyte at The Knoll also proved to be of reversed polarity. Age determinations [*Armstrong*, 1978] on pyroxene trachyte of The Knoll and hornblende trachyte in the vicinity of Post Office Hill have yielded ages of 1.32 ± 0.05 and 1.35 ± 0.04 Ma, respectively. Armstrong also reports an age of 0.8 ± 0.1 Ma for a sample from an olivine basalt flow of the Melania basalt on the cliffs approximately 1 km southeast of the Discovery II Pole. Most of the sampled flows in this area probably erupted during the latter part of the Matuyama Chron with the possible exception of that at site 14. The preponderance of reversed-polarity flows, along with the calcu-

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

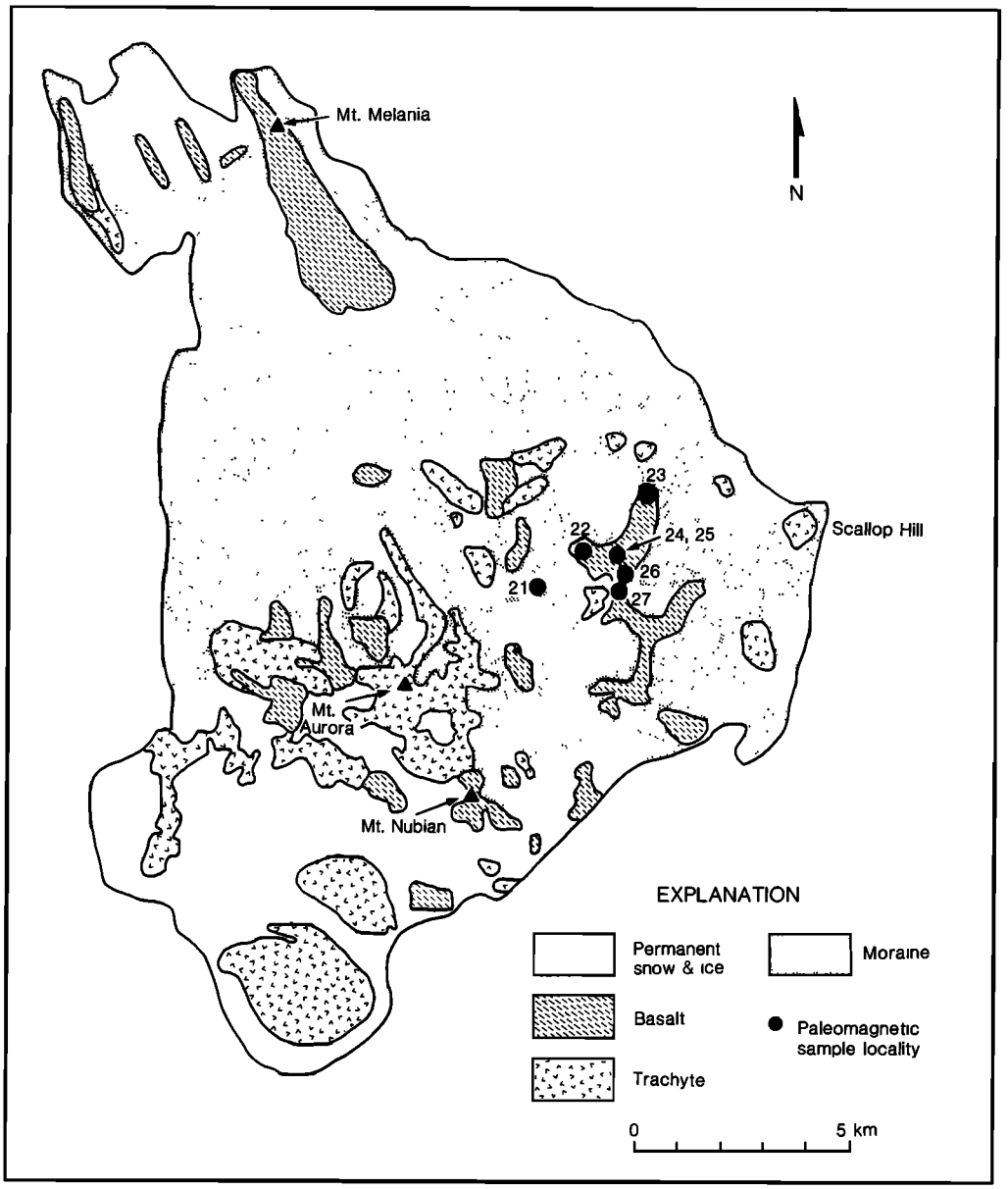


Fig. 5. Map of Black Island showing location of volcanic units sampled. Outcrop distribution and petrology modified from *Cole and Ewart* [1968].

lated ages, suggests that volcanism at Cape Crozier may be slightly older than that at Capes Evans and Royds, and on Hut Point Peninsula.

Dailey Islands (Sites 19 and 20)

Five small cinder cones of the Dailey Islands group protrude through the Ross Ice Shelf approximately 35 km southwest of Hut Point Peninsula (Figure 1). Although the islands are composed mainly of scoria and retain much of their original form, they have been glaciated, as attested to by the occurrence of erratics on their slopes and summits. The erratics here include typical mainland rocks along with numerous boulders and cobbles of kenyte. Sufficient material has been scoured away on some of the cones to reveal several vesicular dikelike bodies. These dikes were sampled on the easternmost of the two northern cones (site 19) and, toward the southwest, on the cone immediately east of the largest Daily Island (site 20). Both cones are normally magnetized, and although their directions are statistically different, they are similar enough to have erupted during the same episode of volcanic activity. No absolute ages are available for these islands, but *Péwé* [1960] considers the erratics on the smallest Dailey Island to have been deposited during the Koettlitz glaciation, which was the last major glaciation in the region. *Cole et al.* [1971] consider the Koettlitz glaciation to be equivalent to the Ross Sea glaciation 1 of *Denton et al.* [1970] which is dated as having occurred between 34.8 and 49.0 kyr.

Black Island (Sites 21–27)

A number of basalt flows and cinder cones surround what *Cole and Ewart* [1968] refer to as the "central plateau" of Black Island (Figure 5). Only two flows were sampled in detail because it was apparent that the volcanics in this area were considerably older than those on Ross Island. Although much

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

scoria remains on the sides of some of the cones, many features are subdued and all flows have been glaciated. The rocks at many localities also have the dark, glossy surfaces that are characteristic of desert polish, presumably resulting from prolonged exposure. Our locality 21 may be in part of what *Cole and Ewart* [1968] mapped as the Nubian basalt, whereas all others are in the Melania basalt. *Armstrong* [1978] reported an age of 11.2 ± 0.4 Ma for a sample from the Melania basalt and much younger ages for samples from the stratigraphically younger Aurora trachyte (3.9 ± 0.1 Ma) and Nubian basalt (3.44 ± 0.14 Ma). The sample number of the rock yielding the oldest age corresponds to one shown in Figure 9 of *Cole and Ewart* [1968] indicating that it came from the volcanic center at the north end of Black Island. *Cole and Ewart* [1968] consider the basalts surrounding the central plateau to be younger than those at the north end of the island. It is possible therefore that the basalt flows we sampled could have erupted during the Gauss Normal-Polarity and Gilbert Reversed-Polarity chrons, but probably no later. Nine samples of red, scoriaceous basalt at site 22 yielded unusually shallow magnetic inclinations when measured with a flux gate magnetometer, but the directions showed considerable scatter because of low magnetic intensities. Two oriented hand samples measured in the laboratory confirmed that this flow has an intermediate polarity direction (Table 1) which, when considered with its very low intensity, indicates that it may have erupted during a polarity transition.

Mount Discovery (Sites 28–30)

Paleomagnetic samples were obtained from three sites on the northwestern flank of Mount Discovery (Figure 6). A youthful appearing basalt flow that forms a small peninsula near the edge of the ice shelf was sampled at site 28. Although a few small patches of morainal material occur on the surface of this flow, it clearly postdates the major glacial advances in the area. Also relatively young is the basaltic dike (site 29) sampled within a small cone at an elevation of approximately 270 m. A few erratics were found on the cone, but they were much less abundant and petrographically less varied than those at the same level on the surrounding slopes. The rocks at both localities proved to be of normal polarity. Older than these two units is the basalt flow (site 30) occurring within a cone at the 1245-m elevation. Erratics are generally absent from this cone, as they are from the surrounding slopes, but the outer rim of the cone has been removed, presumably by glacial action. This flow is reversely magnetized. A feldspar separate from basalt at the summit of Mount Discovery has yielded an age of 5.3 ± 0.1 Ma that *Armstrong* [1978] considers to be suggestive of excess radiogenic argon. No additional age determinations are available to help constrain the age range of volcanism on Mount Discovery. The basalts at our two normal-polarity sites, however, most likely erupted during the Brunhes Chron based on their geomorphic expression.

Mount Morning (sites 31–44)

Paleomagnetic samples were obtained from 14 sites on the northeast flank of Mount Morning (Figure 6) which appears to be the most youthful volcanic field that we investigated. Flows at 13 of these sites were normally magnetized, and their remanence directions are shown in Figure 7. The clustering of directions indicates that eruptions here were not randomly distributed in time, although this impression is not entirely correct because the flows brackets at least one major glacial advance in this area. Because most of the units samples occur at elevations higher than that of the highest moraine (maximum 525 m), determining relationships between eruption and glaciation often was not possible. The highest moraine present consists mainly of granitic rocks with some sandstone and metamorphic rocks. Many of the granitic boulders are deeply weathered and other show evidence of prolonged wind erosion. This moraine was possibly deposited during Taylor glaciation (T. L Péwé, written communication to A. Cox, 1967). One of the older flows (sampled at site 44) near the northern end of the volcanic field has undergone extreme wind erosion and is overlain by the high moraine just described. Northeast of this flow, a thick, columnar-jointed basalt flow was sampled at the base of a 20- to 30-m-high cliff (site 31). Large blocks of this flow (several hundred meters long and about 40 m wide) have slumped away from the main exposure forming the cliff where the samples were taken. Many erratics occur adjacent to this flow but do not overlie it and do not occur in the troughs between the foundered blocks, indicating that the flow is younger than the high moraine. Another columnar-jointed flow (site 33) southeast of site 31 also is thought to be younger than the moraine because it contains abundant granite xenoliths up to 15 cm in diameter. A younger basalt flow (sampled at site 32) that appears to have erupted from a small cone at the 660-m elevation flowed northeastward and over the cliff at site 31 and continued toward the present ice margin. This flow is overlain by a younger moraine at an elevation of 300 m. This younger moraine contains little weathered boulders of granite along with large boulders of basalt that appear to have come from the immediate area. Another of the younger flows in the vicinity (which was not sampled paleomagnetically) overlies the flow sampled at site 44 and the high moraine. This flow yields a whole rock age of 0.28 ± 0.04 Ma (G. B. Dalrymple, written communication, 1967) which attests to the recency of volcanism in this area.

A pahoehoe flow sampled at site 34 appears to have issued from the central vent area of a cone where a basalt flow forming the south rim of the crater was sampled (site 35). The two flows may therefore be equivalent in age, and the remanence directions were found to be statistically the same. The flows sampled at sites 31, 39, and 43 yield remanence directions that are statistically indistinguishable and may indicate eruption at about the same time. The flow at site 31 (400-m elevation) is the thick, columnar-jointed olivine basalt described above, that at site 43 (700-m elevation) is an olivine-bearing pahoehoe flow with well-preserved flow channels, and at site 39 (900-m elevation) is a basalt containing large feldspar phenocrysts that issued from an approximately 150-m-diameter breccia cone. Unless these flows recorded the same geomagnetic field direction by chance, different vents in this volcanic field may have been active at about the same time. The remanence directions at sites 34 and 35 are quite similar to those at sites 31, 39, and 43, but the flows at the latter sites may be considerably younger based on the lesser amount of wind erosion and poorly developed frost-crack polygons on their surfaces.

The oldest flow sampled on the eastern side of the exposed volcanic field (site 37) occurs in a partly denuded breccia cone at the 1000-m elevation. This flow has a reversed magnetic polarity. In general, however, the older units appear to be concentrated near the western part of the field. The terrain in this area is more subdued than farther to the east, and a

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

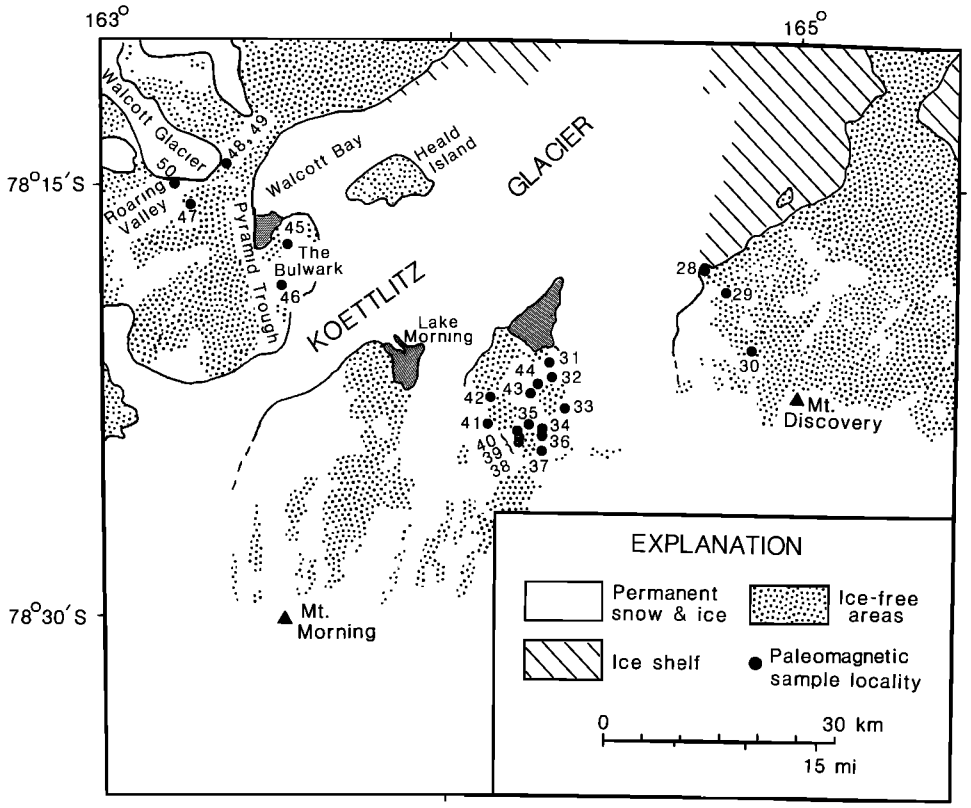


Fig. 6. Map showing paleomagnetic sample localities in the vicinity of Koettlitz Glacier on the Antarctic mainland.

number of partly dissected cones are present. The flow in one of these cones (site 42) proved to be of normal polarity, and that in another was reversed on the basis of fluxgate measurements. The normal-polarity olivine basalt flow sampled at site 41 probably is older than most, if not all, of the flows to the east based on the well-developed frost polygons on its surface.

The lava flows that we sampled on Mount Morning are among the youngest in the McMurdo volcanic province, but this volcanic field was apparently formed over the remnants of a much older volcanic edifice. Trachyandesite flows and breccias that are intruded by a series of peralkaline trachyte dikes were found at the lower elevations on the northern slopes of Mount Morning by *Kyle and Muncy* [1978], who considered the dikes to be feeders to a now eroded peralkaline trachyte volcano. A sample from one of the dikes yielded an age of 15.4 ± 0.5 Ma [*Armstrong*, 1978] which makes it the oldest of the volcanic rocks in the province known to date. Major and trace element analyses by *Kyle and Muncy* [1978], however, show that these rocks are petrographically distinct from the younger basanite-phonolite suite in the region. The only other absolute age information for Mount Morning is a 1.18 ± 0.02 Ma age determined for a sample of trachyte from the upper slopes of the volcano [*Armstrong*, 1978].

Walcott Bay (Sites 45–50)

Areas in the immediate vicinity of Walcott Bay (Figure 6) also were visited during this investigation. At the south end of Walcott Bay, flows and cones covering Dromedary Platform and The Bulwark are truncated abruptly at the walls of Pyramid Trough [*Blank et al.*, 1963]. Large patches of erratics overlie a volcanic breccia (site 45) containing thin flow units at the north end of The Bulwark. This unit is reversely magnetized as is a flow (site 46) from a younger appearing cone at the south end of The Bulwark. *Armstrong* [1978] reports a reversed polarity and an age of 1.7 ± 0.4 Ma for a bomb from the northern summit of The Bulwark near our site 45. A flow at the top of The Bulwark yields an age of 1.69 ± 0.3 Ma, and a lava tongue flowing off The Bulwark into Pyramid Trough farther to the south yields a reversed polarity and an age of 1.88 ± 0.09 Ma [*Armstrong*, 1978]. These three units and a reversely magnetized flow (2.96 ± 0.15 Ma) on the side of The Bulwark indicated to *Armstrong* [1978] that glacial erosion of Pyramid Trough occurred about 2 m.y. ago. Armstrong also reported an average age of 0.2 ± 0.1 Ma and normal polarity for basalt domes on the floor of Pyramid Trough that show evidence of a more recent glaciation, perhaps one of the Ross Sea advances.

A basalt flow occupying the floor of a small cirque on the east side of Roaring Valley (Figure 6) was sampled at site 47. This basalt flowed down the walls of the glacially cut valley and clearly overlies the moraines just below the cirque. No absolute age is available for this flow, but its weak magnetic intensity and intermediate-polarity direction (Table 1) indicate eruption during an excursion or polarity transition. Two lavas that flowed down the side of Roaring Valley southwest of locality 47 have yielded ages of 2.16 ± 0.09 and 1.83 ± 0.19 Ma [*Armstrong*, 1978].

Meltwater emerging from the terminus of Walcott Glacier has cut a small canyon that exposes several basalt flows that are overlain by moraine. The youngest basalt flow sampled in the canyon (site 48) overlies a layer of till containing large boulders of granitic and metamorphic rocks. The flow is, in turn, overlain by a younger moraine of similar character. The flow at site 48 is normally magnetized and is probably the

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

same normal-polarity flow sampled by *Armstrong* [1978] at two sites in the canyon. Samples from these two sites have yielded ages of 0.46 ± 0.07 and 0.28 ± 0.10 Ma, and Armstrong considers this flow to predate Alpine II glaciation [*Denton et al.*, 1970]. Underlying the normal-polarity flow and an older moraine is a columnar-jointed, olivine basalt flow which proved to be reversely magnetized (site 49). *Armstrong* [1978] also reports an underlying, reversed-polarity flow that yielded an age of 1.49 ± 0.15 Ma. Another reversed-polarity flow sampled in the canyon by Armstrong gave an anomalously old age of 5.7 ± 0.5 Ma.

One of the olivine basalt flows that cover extensive moraines on the south side of Walcott Glacier was sampled at site 50. Although younger than the major glaciations in the area, these flows are overlain by scattered patches of erratics that attest to their being covered during one of the younger glacial advances. The flow at this site is reversely magnetized. *Armstrong* [1978] reports ages of 0.92 ± 0.09 and 0.86 ± 0.07 Ma for two of the flows partially blocking the mouth of Roaring Valley near locality 50.

DISCUSSION

The sequences of volcanic rocks of the McMurdo volcanic province that were studied paleomagnetically are compared to the geomagnetic polarity time scale in Figure 8. The comparisons rely heavily upon the geologic and geochronologic studies that have been referred to in the previous sections. Although an exhaustive paleomagnetic sampling was not done in most areas, the results help to establish a general period for the volcanic eruptions and attest to the reliability of the published age determinations. As can be seen in Figure 8, the volcanic rocks in the dry valley areas of the mainland are among the oldest within this province. No large volcanic accumulations formed in this area and eruptions occurred sporadically over long periods of time. This mode of eruption, however, resulted in volcanic deposits alternating with sediments

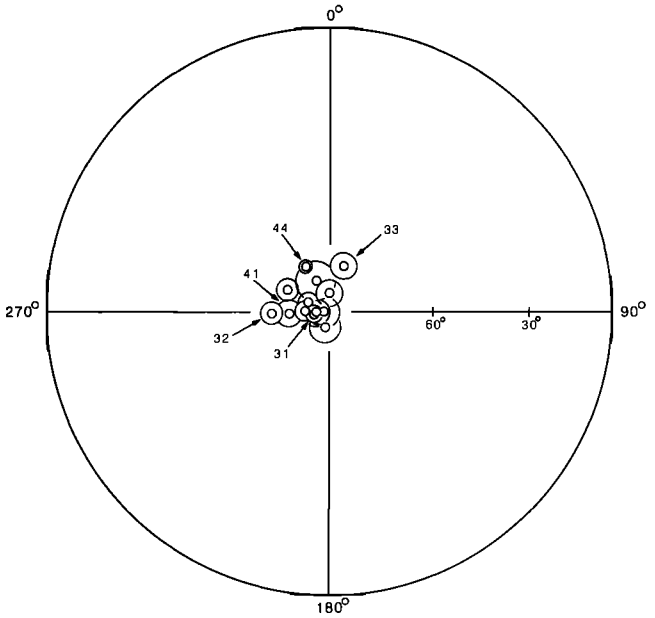


Fig. 7. Average remanent magnetization directions for normal-polarity units sampled on the northeast flank of Mount Morning. See Figure 3 for additional explanation.

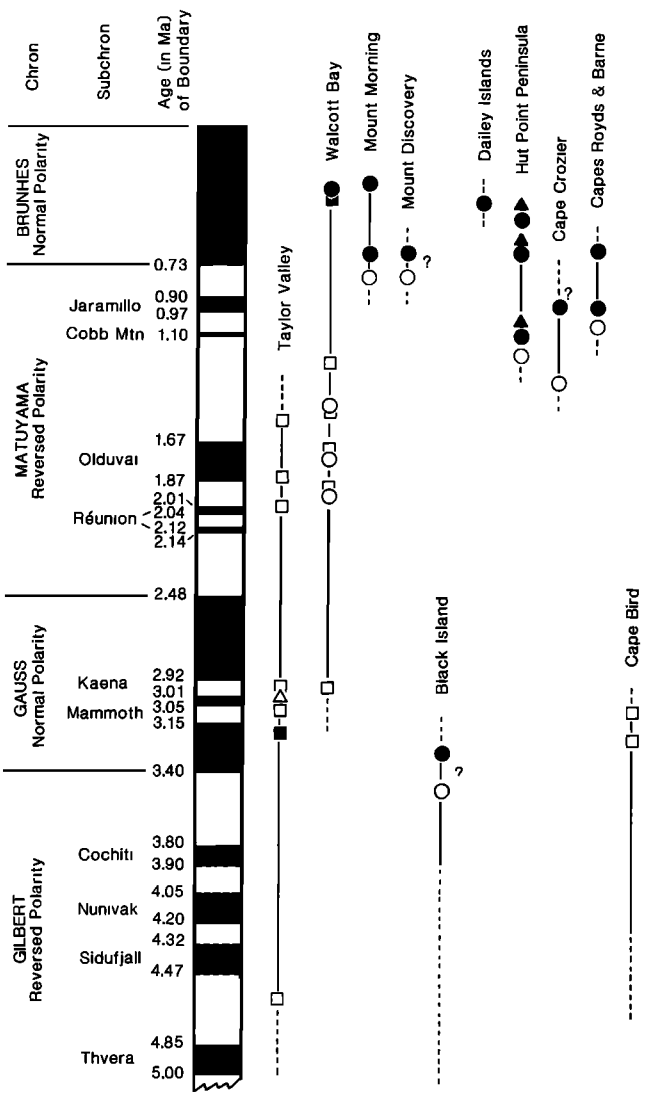


Fig. 8. Correlation of paleomagnetically sampled volcanic units in the McMurdo Sound area with a geomagnetic polarity time scale modified from *Mankinen and Dalrymple* [1979] and including the Cobb Mountain Normal-Polarity Subchron [*Mankinen et al.*, 1978; *Mankinen and Grommé*, 1982]. Circles, data from this study; squares, data from *Armstrong* [1978]; triangles, data from *Funaki* [1984]. Solid symbols, normal polarity; open symbols, reversed polarity. Location of symbols on diagram is intended to show best estimate as to their general age. The most uncertain assignments are denoted by a question mark.

and glacial features which allowed a glacial chronology to be established. Available age determinations show that the dry valleys were carved by throughgoing glaciers prior to 4 Ma and have experienced repeated glaciation since that time (see summary by *Armstrong* [1978] for a detailed description). Eruptions apparently began during the Gilbert Reversed-Polarity Chron and a few continued into the Brunhes Chron. The most recent eruptions seem to be the small scoria mounds that occur on the floor of Pyramid Trough [*Blank et al.*, 1963; *Armstrong*, 1978]. Similar mounds in Roaring Valley yield ages of about 1.2 Ma [*Armstrong*, 1978].

Contemporaneous with, or slightly older than the volcanic rocks in the dry valley areas are the flows and cones that were sampled on Black Island. These younger units apparently erupted over the remnants of a deeply eroded stratovolcano

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[*Hamilton*, 1965; *Cole and Ewart*, 1968]. Eruption of the units sampled by us probably spanned the boundary between the Gilbert and Gauss chrons, although much older volcanism occurred at the north end of the island near Mount Melania. We did not take samples from White Island, Brown Peninsula, or Minna Bluff, all of which are topographically similar to Black Island. Similar volcanic sequences occur on Black Island and Brown Peninsula [*Cole and Ewart*, 1968; *Kyle et al.*, 1979] but are slightly younger (2.7–2.1 Ma) on Brown Peninsula [*Armstrong*, 1978]. The geology of White Island was described by *Cole et al.* [1971] and of Minna Bluff by *Wright et al.* [1983], but no absolute age determinations are available for either area. The youngest volcanism in the immediate area was on Mount Morning where eruptions probably began during the Matuyama Chron and extended well into the Brunhes. Mount Discovery is topographically very young, and volcanism must have been contemporaneous, at least in part, with that on Mount Morning, although there is still little known about Mount Discovery.

Between the mainland and Ross Island, the cinder cones composing the Dailey Islands probably erupted late in the Brunhes. The youngest volcanism in the province, however, occurred on Ross Island where the historically active Mount Erebus began its latest period of eruptive activity prior to 1973. A convective lava lake in the crater of Mount Erebus was discovered in January 1973, and strong explosive activity began in 1984 [*Kienle et al.*, 1985]. The oldest part of Ross Island is the area around Mount Bird at the northern end of the island. We did not visit this area because of logistic difficulties, but *Armstrong* [1978] has obtained reversed polarities on two samples of trachyte that yield ages of 3.0 ± 0.2 and 3.23 ± 0.09 Ma, possibly indicating eruption during the Mammoth Reversed-Polarity Subchron. Armstrong also reported ages of 4.5 ± 0.6 and 3.7 ± 0.2 Ma for two other units in the area. A geologic description is provided by *Cole and Ewart* [1968]. Our sampling and age determinations by *Armstrong* [1978] show that eruptions near Cape Crozier occurred mainly during the Matuyama Chron and partially overlapped the volcanic activity at Hut Point Peninsula, Cape Royds, and Cape Barnes. Activity at the latter areas continued into the Brunhes. We were unable also to visit or sample the most recent products from Mount Erebus.

The VGPs of all units that we sampled in detail (Table 1) are shown in Figure 9. These 39 VGPs were averaged and they yielded a paleomagnetic pole at 86.6°N, 334.8°E, with $R = 36.0873$, $k = 13$, and $\alpha_{95} = 6.6°$. VGPs calculated for the 14 units sampled by *Funaki* [1984] were said to center near the geographic south pole, although a mean pole position was not calculated. Eight of the VGPs that were based on multiple samples and had α_{95} less than 10° [*Funaki*, 1984] were combined with our data to yield a mean paleomagnetic pole at 87.3°N, 317.3°E, with $R = 43.1753$, $k = 12$, and $\alpha_{95} = 6.3°$. Funaki's Fortress Rock and Royds A sites were excluded because they probably duplicate our sampling. Data with which these results can be compared have been reported from the Hallett volcanic province in northeastern Victoria Land and from the Cenozoic volcanics of Marie Byrd Land. Single oriented samples were obtained from 23 Cenozoic lava flows in the vicinity of Cape Hallett by *Turnbull* [1959], who found that 21 of these were of reversed polarity. These samples yielded a mean VGP at 81°S, 94°E, with semiaxes of the 95% confidence oval equal to 7.8° and 8.1°. *Delisle* [1983] obtained

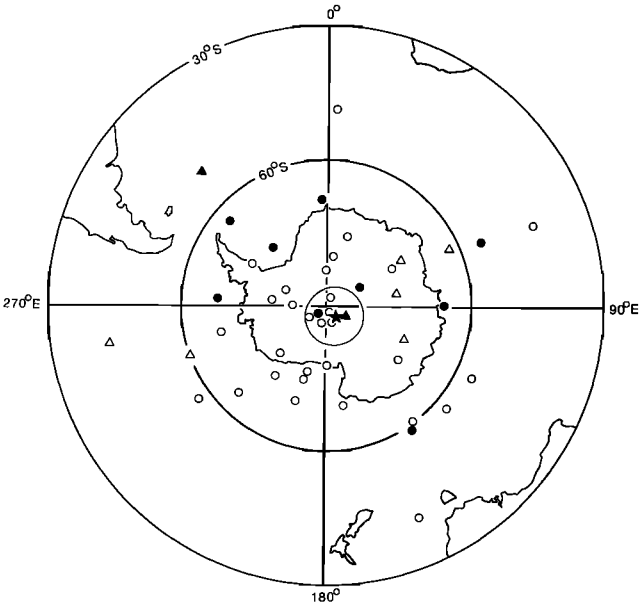


Fig. 9. Equal-area projection showing virtual geomagnetic pole (VGP) positions of volcanic units from the McMurdo volcanic province. Solid symbols, reversed polarity; open symbols, normal-polarity VGPs inverted through the origin. Circles, data from this study; triangles, data from *Funaki* [1984]. Star, mean paleomagnetic pole with its associated 95% confidence limit.

two or three oriented cores from four basalt flows on Hallett Peninsula and 2–12 cores from 10 flows on Adare Peninsula. Results from these units yield paleomagnetic poles at 81.9°S, 256.8°E (Cape Hallett) and 83.6°S, 186.4°E (Cape Adare), but both have large associated uncertainties. Many of the volcanic units in the vicinity of Capes Hallett and Adare are probably late Miocene in age [*Armstrong*, 1978; *Delisle*, 1983]. *Scharon et al.* [1969] reported a paleomagnetic pole position of 78°S, 128°W for Pleistocene volcanic rocks from Marie Byrd Land. Details as to number of units sampled or the sampling procedures used were not given, and an inspection of Figure 1 in that report indicates that the uncertainty about the mean pole may be of the order of 12°. Thus studies of Cenozoic volcanic rocks from Antarctica, although still limited in number, do yield similar paleomagnetic pole positions.

Because the volcanic rocks of the McMurdo volcanic province that have been sampled span an appreciable length of time (Figure 8) and because the paleomagnetic pole is within 3° of the modern rotational axis, we conclude that secular variation has been successfully averaged. The total angular dispersion, S_T, of VGPs about the mean paleomagnetic pole is 23.7°, and the dispersion corrected for within-site dispersion due to experimental errors, S_F, is 23.5°. The upper and lower 95% confidence limits for S_F are equal to 27.4° and 20°5°, respectively, as calculated using the method of *Cox* [1969*b*]. Even though this data set is biased toward normal polarity (37 normal sites, 10 reversed), the 23.5° of dispersion that we calculate is probably a reasonable estimate of the Pliocene-Pleistocene secular variation for the Antarctic continent. By way of comparison, paleosecular variation model F of *McFadden and McElhinny* [1984] predicts a dispersion of 20.6° at the latitude of our study area during the last 5 m.y. Although confidence limits cannot be determined for the values obtained

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

from their figures, it appears that there is little or no significance to the difference between the predicted and measured dispersions.

The value for dispersion of Antarctic VGPs given by *Doell* [1972*c*], *Doell and Cox* [1971, 1972], and *Doell and Dalrymple* [1973] was based on those flows that were originally considered to of Brunhes age. This value can now be recalculated as can be the value for flows thought to represent the Matuyama Chron as given by *Doell and Dalrymple* [1973]. The original calculation for Brunhes dispersion included data for the flow at site 26 on Black Island that probably erupted during the Gauss Chron and the two flows at Cape Barne (sites 10 and 11) that probably erupted during the Jaramillo Subchron. Data from the Crater Hill sequence lavas (sites 2 and 3) probably should be excluded because of the uncertainty as to their age. Data from *Funaki*'s [1984] Scott Hut, Second Crater, Black Knob, and Royds B sites can also be added to this data set. These 27 data yield 19.3° of dispersion, with upper and lower confidence limits of 23.5° and 15.9°. Of the nine data originally used for calculating the dispersion during the Matuyama [*Doell and Dalrymple*, 1973], only those for site 21 on Black Island should be excluded. To this group can be added the normal-polarity data from the two sites at Cape Barne (Jaramillo Subchron), the dike at Castle Rock (site 7) and *Funaki*'s [1984] Half Moon Crater site (both Jaramillo or Cobb Mountain Subchron), and Funaki's reversed-polarity Observation Hill site. These 13 data yield 26.5° of dispersion with upper and lower confidence limits of 36.3° and 20.4°. The value for dispersion during the Brunhes is the same as the earlier estimate, whereas the value for the Matuyama is 2.5° larger. This latter difference probably is not meaningful because the data are still few in number and the low-latitude VGPs for the Castle Rock–Half Moon Crater units may or may not be entirely representative of normal secular variation.

Acknowledgments. Special thanks are extended to R. C. Wood and W. B. Emison of Johns Hopkins University for their hospitality during our stay at Cape Crozier, to U. Kinet of the Bishop Museum for companionship in the field, and to the helicopter pilots and crews of U.S. Navy Squadron VXE-6 for their support of our field operations. Constructive reviews of this manuscript provided by S. W. Bogue, A. B. Ford, C. S. Grommé, W. P. Harbert, and one anonymous reviewer are also appreciated. The first author wishes to acknowledge Allan Cox whose enthusiasm for paleomagnetism and volcanology helped influence my decision to pursue paleomagnetic research.

REFERENCES

Armstrong, R. L., K-Ar dating—Late Cenozoic McMurdo Volcanic Group and dry valley glacial history, Victoria Land, Antarctica, *N. Z. J. Geol. Geophys.*, *21*, 685–698, 1978.

Armstrong, R. L., W. Hamilton, and G. H. Denton, Glaciation in Taylor Valley, Antarctica, older than 2.7 million years, *Science*, *159*, 187–189, 1968.

Blank, H. R., R. A. Cooper, R. H. Wheeler, and I. A. G. Willis, Geology of the Koettlitz-Blue Glacier region, Southern Victoria Land, Antarctica, *Trans. R. Soc. N. Z.*, *2*, 79–100, 1963.

Boudette, E. L., and A. B. Ford, Physical properties of anorthoclase from Antarctica, *Am. Mineral.*, *51*, 1374–1387, 1966.

Cole, J. W., and A. Ewart, Contributions to the volcanic geology of the Black Island, Brown Peninsula, and Cape Bird areas, McMurdo Sound, Antarctica, *N. Z. J. Geol. Geophys.*, *11*, 793–828, 1968.

Cole, J. W., P. R. Kyle, and V. E. Neall, Contributions to Quaternary geology of Cape Crozier, White Island, and Hut Point Peninsula, McMurdo Sound region, Antarctica, *N. Z. J. Geol. Geophys.*, *14*, 528–546, 1971.

Cox, A., Paleomagnetic research on volcanic rocks of McMurdo Sound, *Antarct. J. U.S.*, *1*, 136, 1966.

Cox, A., A paleomagnetic study of secular variation in New Zealand, *Earth Planet. Sci. Lett.*, *6*, 257–267, 1969*a*.

Cox, A., Confidence limits for the precision parameter *k*, *Geophys. J. R. Astron. Soc.*, *18*, 545–549, 1969*b*.

Cox, A., Paleomagnetism of San Cristobal Island, Galapagos, *Earth Planet. Sci. Lett.*, *11*, 152–160, 1971.

Cox, A., and R. R. Doell, Long period variations of the geomagnetic field, *Bull. Seismol. Soc. Am.*, *54*, 2243–2270, 1964.

Debenham, F., Recent and local deposits of the McMurdo Sound region: British Antarctic ("Terra Nova") Expedition, 1910, *Nat. Hist. Rep. Geol.*, *1*, 63-100, 1921.

Delisle, G., Results of paleomagnetic investigations in northern Victoria Land, Antarctica, in *Antarctic Earth Science*, edited by R. L. Oliver, P. R. James, and J. B. Jago, pp. 146–149, Cambridge University Press, Cambridge, 1983.

Denton, G. H., R. L. Armstrong, and M. Stuiver, Late Cenozoic glaciation in Antarctica: The record in the McMurdo Sound region, *Antarct. J. U.S.*, *5*, 15-21, 1970.

Doell, R. R., Paleomagnetism of the Kau volcanic series, Hawaii, *J. Geophys. Res.*, *74*, 4857–4868, 1969.

Doell, R. R., Paleomagnetic secular variation study of lavas from the Massif Central, France, *Earth Planet. Sci. Lett.*, *8*, 352–362, 1970.

Doell, R. R., Paleomagnetism of lava flows from Kauai, Hawaii, *J. Geophys. Res.*, *77*, 862–876, 1972*a*.

Doell, R. R., Paleosecular variation of the Honolulu Volcanic Series, Oahu, Hawaii, *J. Geophys. Res.*, *77*, 2129–2138, 1972*b*.

Doell, R. R., Palaeomagnetic studies of Icelandic lava flows, *Geophys. J. R. Astron. Soc.*, *26*, 459–479, 1972*c*.

Doell, R. R., and A. Cox, Paleomagnetism of Hawaiian lava flows, *J. Geophys. Res.*, *70*, 3377–3405, 1965.

Doell, R. R., and A. Cox, Pacific geomagnetic secular variation, *Science*, *171*, 248–254, 1971.

Doell, R. R., and A. Cox, The Pacific geomagnetic secular variation anomaly and the question of lateral uniformity in the lower mantle, in *The Nature of the Solid Earth*, edited by E. C. Robertson, J. F. Hays, and L. Knopoff, pp. 245–284, McGraw-Hill, New York, 1972.

Doell, R. R., and G. B. Dalrymple, Potassium-argon ages and paleomagnetism of the Waianae and Koolau Volcanic Series, Oahu, Hawaii, *Geol. Soc. Am. Bull.*, *84*, 1217–1242, 1973.

Fleck, R. J., L. M. Jones, and R. E. Behling, K-Ar dates of the McMurdo Volcanics and their relation to the glacial history of Wright Valley, *Antarct. J. U.S.*, *7*, 244–246, 1972.

Forbes, R. B., D. L. Turner, and J. R. Carden, Age of trachyte from Ross Island, Antarctica, *Geology*, *2*, 297–298, 1974.

Funaki, M., The paleomagnetic investigation of Antarctica, 1, Paleomagnetism of Hut Point Peninsula volcanic sequence, *Mem. Natl. Inst. Polar Res., Spec. Issue Jpn.*, *14*, 186–193, 1979.

Funaki, M., Paleomagnetic investigation of McMurdo Sound region, Southern Victoria Land, Antarctica, *Mem. Natl. Inst. Polar Res. Jpn., Ser. C, Earth Sci.*, *16*, 1–81, 1984.

Glen, W., *The Road to Jaramillo*, 459 pp., Stanford University Press, Stanford, 1982.

Hamilton, W., The Ross Sea volcanic province, Antarctica, paper presented at International Symposium on Volcanology, Int. Assoc. of Volcanol., Auckland, New Zealand, 1965.

Hamilton, W., The Hallett volcanic province, Antarctica, *U.S. Geol. Surv. Prof. Pap.*, *456-C*, 62 pp., 1972.

Harrington, H. J., Nomenclature of rock units in the Ross Sea region, Antarctica, *Nature*, *182*, 290, 1958.

Kienle, J., C. A. Rowe, P. R. Kyle, W. C. McIntosh, and R. R. Dibble, Eruption of Mount Erebus and Ross Island seismicity, 1984–1985, *Antarct. J. U.S.*, *19*, 25–28, 1985.

Kyle, P. R., The geology and volcanic history of Hut Point Peninsula as inferred from DVDP 1, 2 and 3 drillcores, *Dry Valley Drill. Proj. Bull. 8*, pp. 39–43, North. Ill. Univ., Dekalb, 1978.

Kyle, P. R., Mineralogy and geochemistry of a basanite to phonolite sequence at Hut Point Peninsula, Antarctica, based on core from Dry Valley Drilling Project drillholes 1, 2, and 3, *J. Petrol.*, *22*, 451–500, 1981.

Kyle, P. R., and H. L. Muncy, Volcanic geology of the lower slopes of Mt. Morning, *Antarct. J. U.S.*, *13*, 34–36, 1978.

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Kyle, P. R., and S. B. Treves, Geology of Hut Point Peninsula, Ross Island, *Antarct. J. U.S.*, *9*, 232–234, 1974*a*.

Kyle, P. R., and S. B. Treves, Geology of DVDP 3, Hut Point Peninsula, Ross Island, Antarctica, *Dry Valley Drill. Proj. Bull. 3*, pp. 13–48, North. Ill. Univ., Dekalb, 1974*b*.

Kyle, P. R., J. F. Sutter, and S. B. Treves, K/Ar age determinations on DVDP 1 and 2 core samples, *Dry Valley Drill. Proj. Bull. 8*, pp. 46–47, North. Ill. Univ., Dekalb, 1978.

Kyle, P R., J. Adams, and P. C. Rankin, Geology and petrology of the McMurdo volcanic group at Rainbow Ridge, Brown Peninsula, Antarctica, *Geol. Soc. Am. Bull.*, *90*, 676–688, 1979.

Mankinen, E. A., and G. B. Dalrymple, Revised geomagnetic polarity time scale for the interval 0–5 m.y. B.P., *J. Geophys. Res.*, *84*, 615–626, 1979.

Mankinen, E. A., and C. S. Grommé, Paleomagnetic data from the Coso Range, California and current status of the Cobb Mountain normal geomagnetic polarity event, *Geophys. Res. Lett.*, *9*, 1279–1282, 1982.

Mankinen, E. A., J. M. Donnelly, and C. S. Grommé, Geomagnetic polarity event recorded at 1.1 m.y. B.P. on Cobb Mountain, Clear Lake volcanic field, California, *Geology*, *6*, 653–656, 1978.

McElhinny, M. W., and R. T. Merrill, Geomagnetic secular variation over the past 5 m.y., *Rev. Geophys.*, *13*, 687–708, 1975.

McFadden, P. L., and D. L. Jones, The fold test in palaeomagnetism, *Geophys. J. R. Astron. Soc.*, *67*, 53–58, 1981.

McFadden, P. L., and M. W. McElhinny, A physical model for palaeosecular variation, *Geophys. J. R. Astron. Soc.*, *78*, 809–830, 1984.

McMahon, B. E., and H. Spall, Paleomagnetic data from unit 13, DVDP hole 2, Ross Island, *Antarct. J. U.S.*, *9*, 229–232, 1974.

Péwé, T. L., Multiple glaciation in the McMurdo Sound region, Antarctica—A progress report, *J. Geol.*, *68*, 498–514, 1960.

Scharon, L., A. Shimoyama, and C. Sharnberger, Paleomagnetic investigations in the Ellsworth Land area, Antarctica, *Antarct. J. U.S.*, *4*, 94–95, 1969.

Smith, W. C., The volcanic rocks of the Ross Archipelago: British Antarctic ("Terra Nova") Expedition, 1910, *Nat. Hist. Rep. Geol.*, *2*, 1–107, 1954.

Steiger, R. H., and E. Jäger, Subcommission on geochronology: Convention on the use of decay constants in geo- and cosmochronology, *Earth Planet. Sci. Lett.*, *36*, 359–362, 1977.

Treves, S. B., The geology of Cape Evans and Cape Royds, Ross Island, Antarctica, in *Antarctic Research: The Matthew Fontaine Maury Memorial Symposium, Geophys. Monogr. Ser.*, vol. 7, edited by H. Wexler, M. J. Rubin, and J. E. Caskey, pp. 40–46, AGU, Washington, D. C., 1962.

Treves, S. B., Volcanic rocks of the Ross Island area, *Antarct. J. U.S.*, *3*, 108–109, 1968.

Treves, S. B., and P. R. Kyle, Geology of DVDP 1 and 2, Hut Point Peninsula, Ross Island, Antarctica, *Dry Valley Drill. Proj. Bull. 2*, pp. 11–82, North. Ill. Univ., Dekalb, 1973.

Turnbull, G., Some paleomagnetic measurements in Antarctica, *Arctic*, *12*, 151–157, 1959.

Wellman, H. W., Later geological history of Hut Point Peninsula, Antarctica, *Trans. R. Soc. N. Z.*, *2*, 147–154, 1964.

Wright, A. C., W. McIntosh, and P. Ellerman, Volcanic geology of Turks Head, Tryggve Point, and Minna Bluff, southern Victoria Land, *Antarct. J. U.S.*, *18*, 35–36, 1983.

E. A. Mankinen, Branch of Isotope Geology, U.S. Geological Survey, 345 Middlefield Road, MS 937, Menlo Park, CA 94025.

(Received September 28, 1987;
revised January 26, 1988;
accepted January 29, 1988.)

21562202b, 1988, B10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB10p11599 by Curtin University Library, Wiley Online Library on [13/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License